FINAL PROJECT REPORT

Predicting Falcon 9 first-stage landings

Evidence-based report from executed Jupyter notebooks

IBM Data Science Professional Certificate | October 2026

GitHub: github.com/hajar-hajiri/IBM-Data-Science-Capstone-SpaceX

Executive summary

Research question: Which mission and flight characteristics are associated with successful Falcon 9 first-stage landings?

IBM classification dataset: 90 Falcon 9 launches (2010–2020); 60 successful landings (66.7%).

Four predictive models evaluated on the same test sample (18 launches). SVM achieved the highest observed test accuracy: 83.3%.

Evidence: executed EDA, SQL queries, Folium HTML export, Plotly chart and confusion-matrix figures.

Business context & research objective

Reusable launch vehicles can reduce the cost of spaceflight by recovering and reusing the first stage.

The project examines historical SpaceX launches to describe performance by site, orbit, payload and time, then trains classifiers to estimate first-stage landing outcomes.

Target Class: 1 = successful landing; 0 = unsuccessful or no landing.

Caution: associations in this observational dataset do not establish causality.

End-to-end methodology

01 Collect launch data: historical API response and Wikipedia scraping

02 Clean features: missing payload values, outcome labels, dates

03 Explore: Pandas, SQL, Seaborn, launch-site maps and interactive dashboard

04 Predict: train/test split, hyperparameter search, model evaluation

05 Interpret: compare hold-out accuracy and note dataset limitations

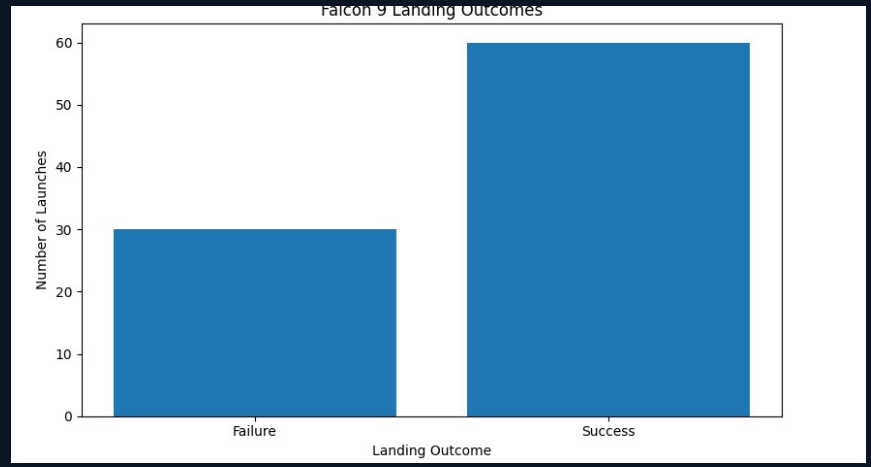
Data collection and provenance

Source	Dataset / evidence	Purpose
IBM API-derived dataset	dataset_part_1.csv (90 × 17)	Launch-level classification features
Wikipedia scraping	spacex_web_scraped.csv (259 × 12)	Historical launch-table extraction
IBM SQL lab	SPACEXTBL query CSV exports	Mission, payload and site aggregates

The SQL laboratory data has a different population and naming conventions from the 90-row classification dataset. Results are reported separately, not pooled.

Data wrangling and target definition

Metric	Verified value
Classification dataset	90 rows; 17 source fields
Missing LandingPad	26 values
Imputed PayloadMass	Mean imputation in wrangling workflow
Class = 1	60 successful landings
Class = 0	30 other outcomes



LandingPad missing values are retained; they may encode missions without an assigned landing pad.

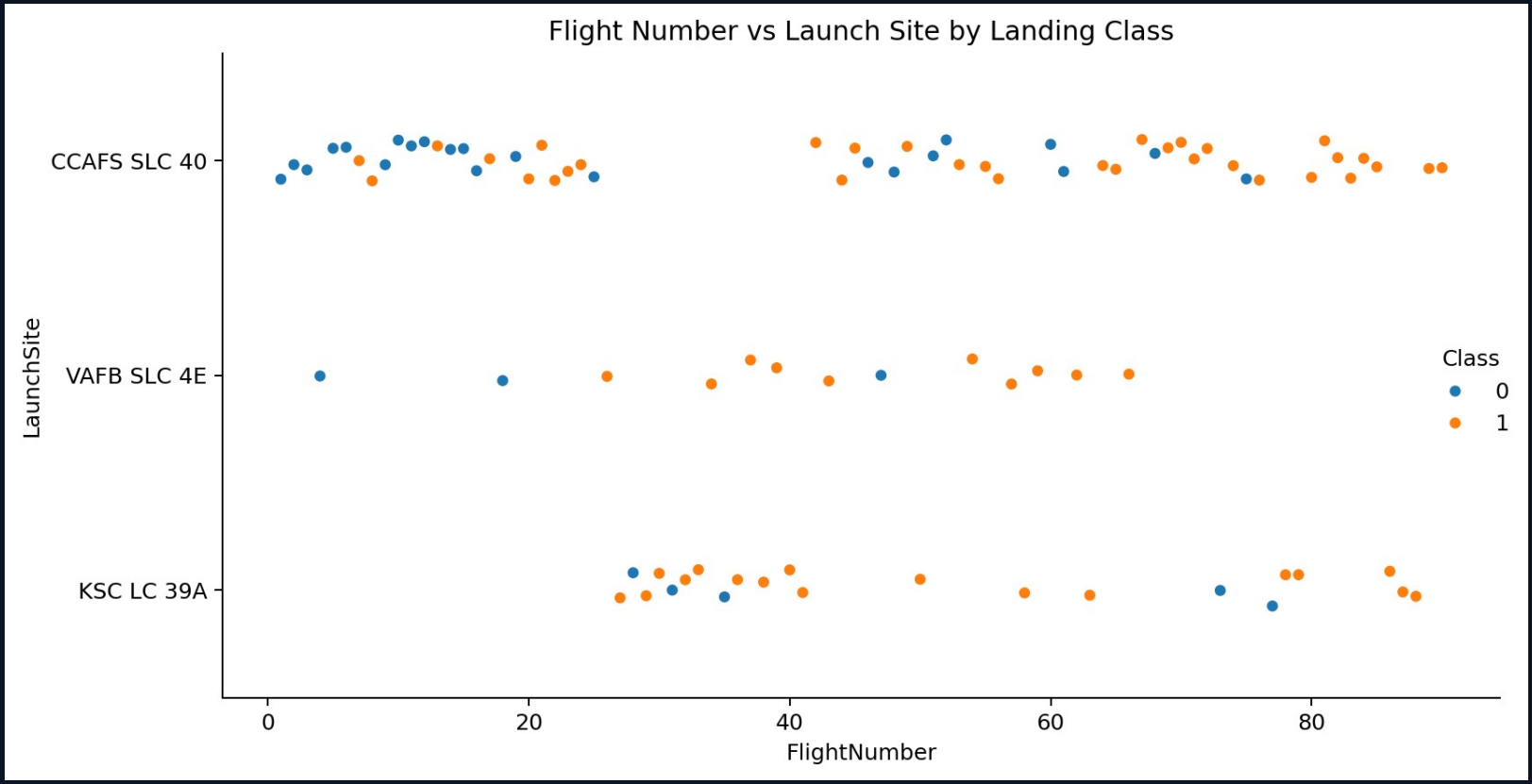
EDA visualization methodology

Question-driven plots link launch characteristics to landing success.

- Scatter / categorical plots: FlightNumber vs LaunchSite, PayloadMass vs LaunchSite and Orbit
- Group-level comparison: landing success across orbits
- Time trend: annual landing success rate

Images on the following slides are taken directly from the uploaded figures/ folder. No performance values have been fabricated.

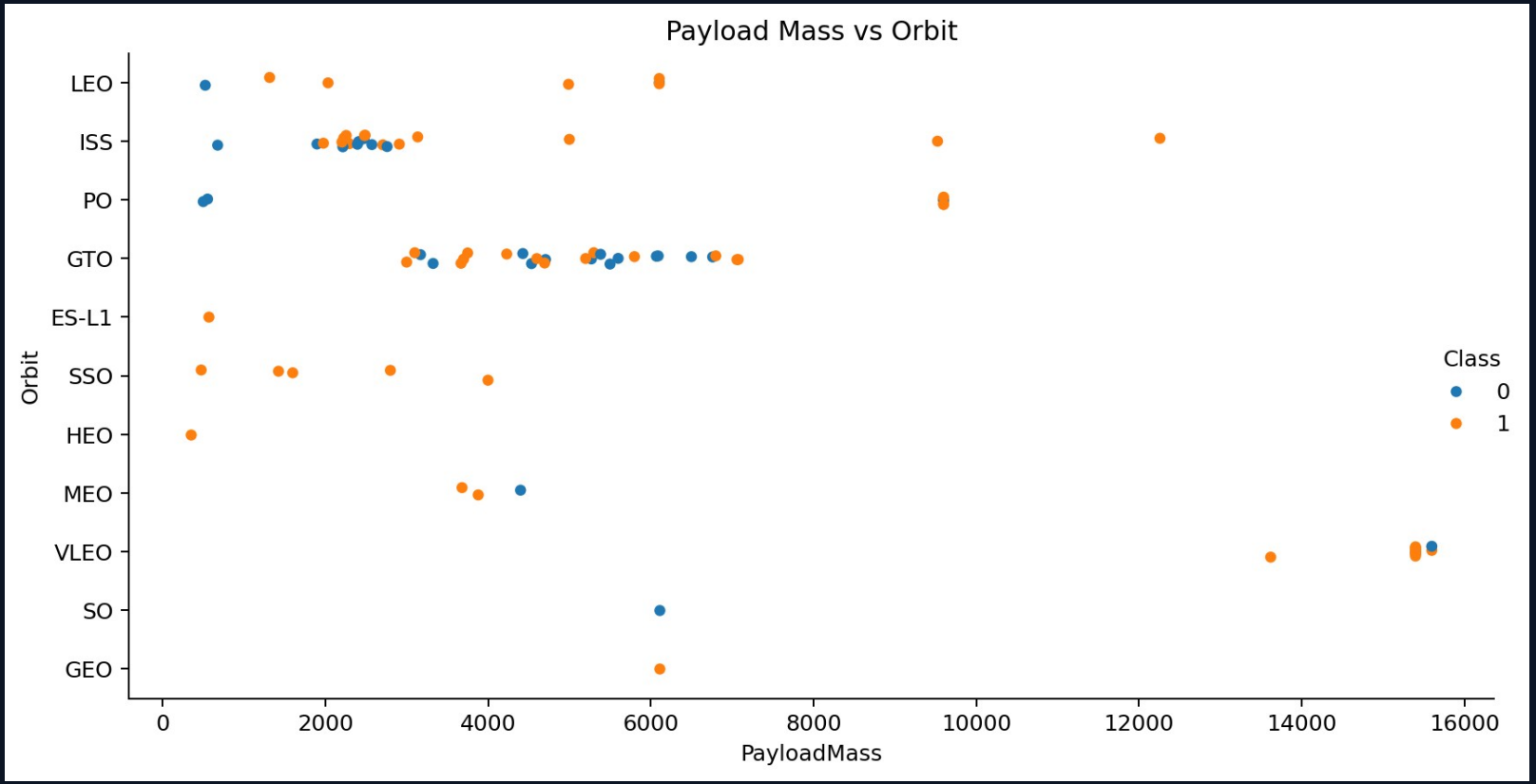
EDA: launch site and flight number



The plot shows how the flight sequence is distributed over launch sites, with outcomes marked by class.

Interpretation: site-specific counts alone cannot explain success; launches occur in different years and mission contexts.

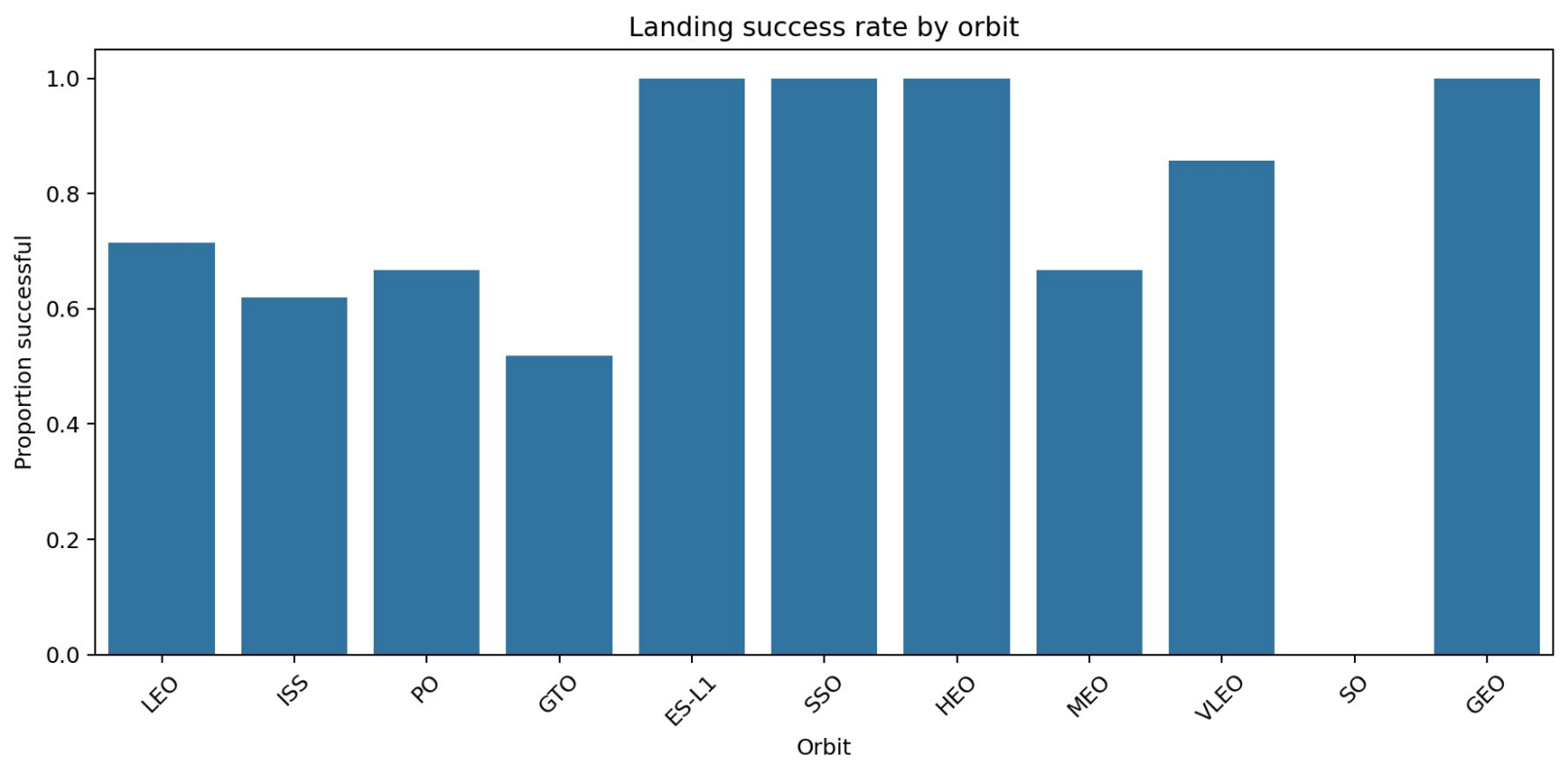
EDA: payload mass and orbit



Payload mass differs across orbit categories.

The observed distributions can affect landing feasibility, but should not be read as a causal effect without controlling for time, booster version and mission profile.

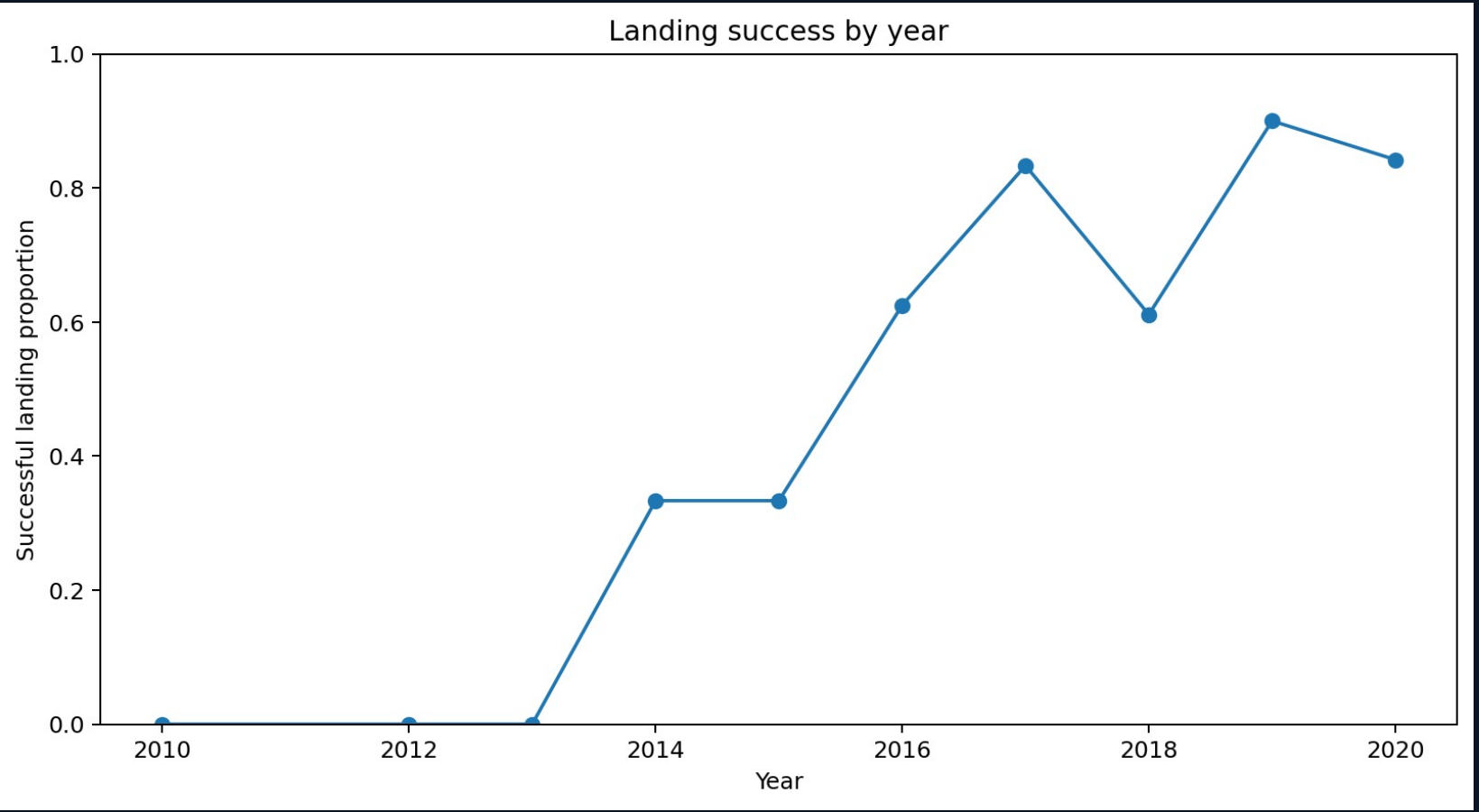
EDA: outcome by orbit



Observed landing outcomes vary by orbit type.

Unequal orbit sample sizes matter: a high rate based on few launches is less stable than a high rate based on many launches.

EDA: temporal evolution



The IBM dataset shows early launch years with few or no recovered stages, followed by substantial improvement.

2019: 90.0% (10 launches).

2020: 84.2% (19 launches).

SQL methodology & source separation

The SQL analysis loads the IBM SpaceX SQL CSV into SQLite (SPACEXTBL), then uses DISTINCT, GROUP BY, COUNT, SUM and aggregate functions.

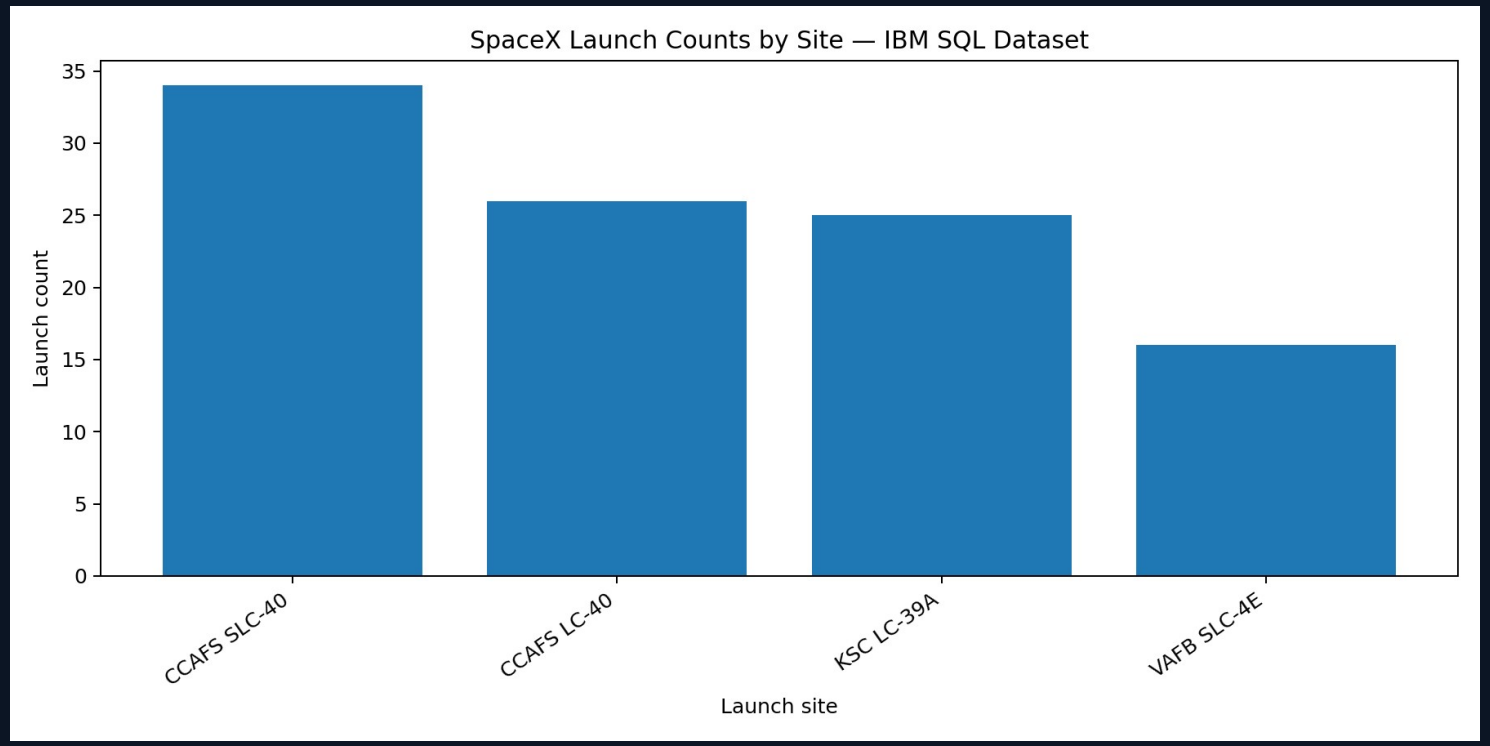
Example SQL queries:

```
SELECT Launch_Site, COUNT(*) AS launches FROM SPACEXTBL GROUP BY Launch_Site;
```

```
SELECT MIN(PAYLOAD_MASS__KG_) AS minimum_payload FROM SPACEXTBL;
```

Outputs below are recorded CSV query results, rather than inferred counts.

SQL: launch sites and mission outcomes



SQL site label	Launches
CCAFS SLC-40	34
CCAFS LC-40	26
KSC LC-39A	25
VAFB SLC-4E	16

SQL counts sum to 101 launches. Site naming differs from the 90-row classification dataset.

SQL: payload and landing statistics

SQL metric	Executed result
Minimum payload (kg)	0
Maximum payload (kg)	15,600
Average payload (kg)	6,138.29
Total payload mass (kg)	619,967
Successful drone-ship landings	14
GEO launches	0
GTO launches	30

GEO and GTO are distinct orbital labels. These values come from SQL exports in data/.

Folium: launch-site geospatial analysis

Folium maps launch-pad latitude/longitude from dataset_part_2.csv and allows interactive inspection of each site.

The uploaded project includes dashboard/launch_sites_map.html, an interactive map export.

A static screenshot of this HTML map is not included in the ZIP; this report documents the file instead of claiming an unavailable screenshot.

Take a browser screenshot of the HTML map to strengthen the Coursera evidence.

Plotly Dash: interactive visualization

Interactive evidence in the uploaded project:

- Notebook 07_Plotly_Dash.ipynb has three executed cells with outputs.
- dashboard/dash_launches_by_site.html contains an interactive Plotly chart.

The chart enables exploration of launch-site launch counts.

A full running Dash application screenshot is not present in the ZIP; an HTML chart is not equivalent to a deployed Dash callback application.

Predictive analysis: experimental design

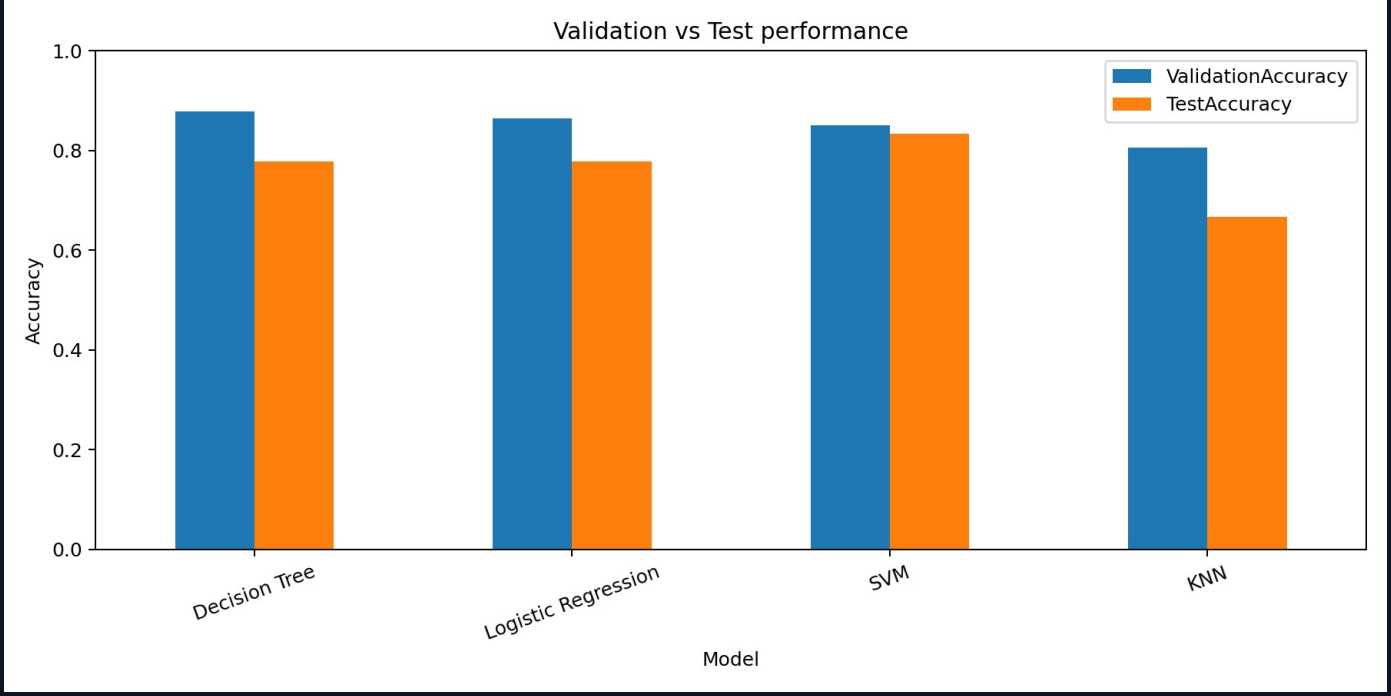
Historical 90-row classification dataset, with Class as the target.

Four fitted classifiers: Logistic Regression, SVM, Decision Tree and KNN.

Best hyperparameters are selected using cross-validation on the training split, then the selected models are evaluated on the 18-record test sample.

A small test sample means accuracy changes in large steps (1 launch = 5.56 percentage points).

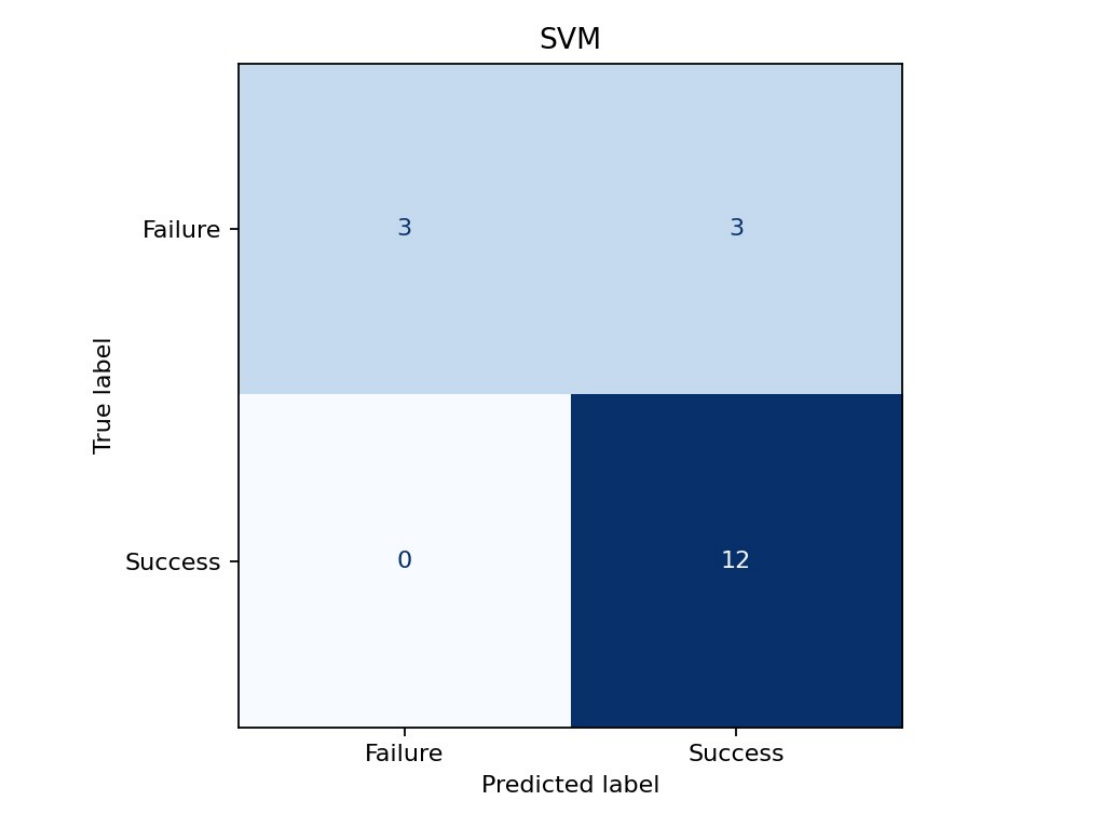
Predictive analysis: observed results

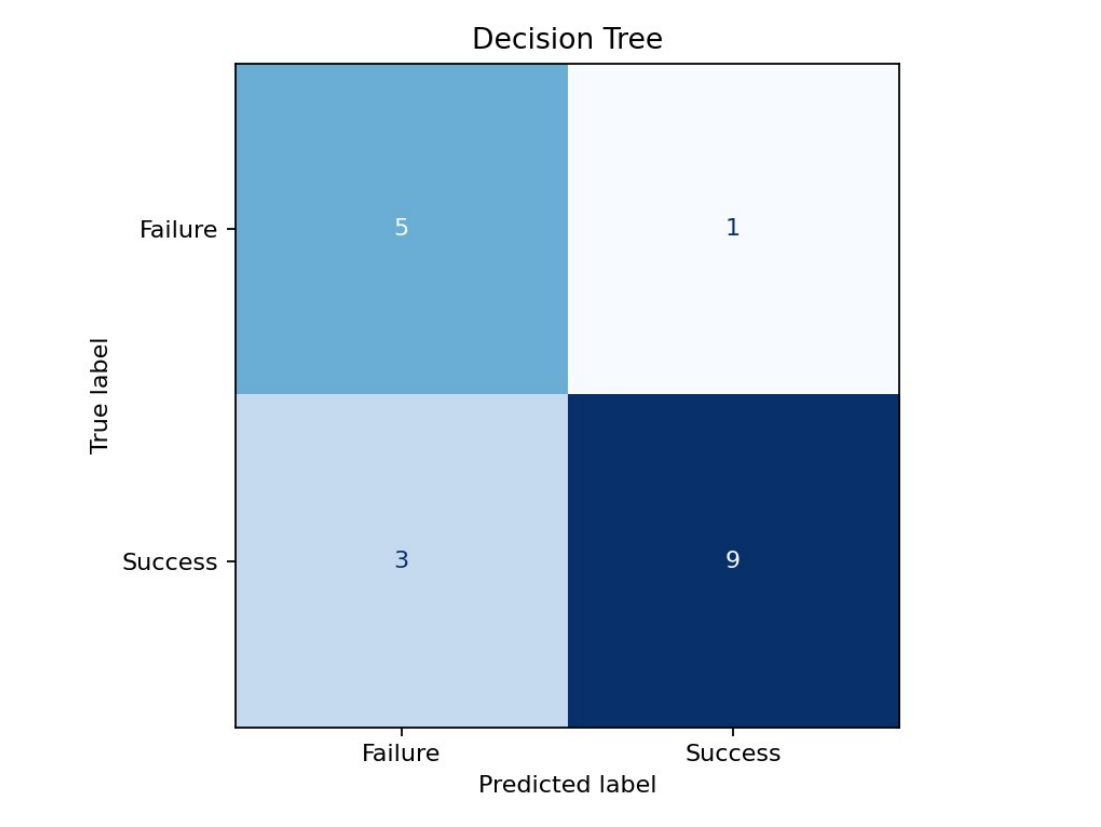


Model	CV accuracy	Test accuracy
Decision Tree	87.8%	77.8%
Logistic Regression	86.4%	77.8%
SVM	85.0%	83.3%
KNN	80.6%	66.7%

SVM recorded the strongest test score in this uploaded run, not the strongest CV mean.

Confusion matrices: model diagnostics





Test-set confusion matrices from notebook 08 outputs; compare false positives and false negatives, not accuracy alone.

Insights, limitations and conclusion

Findings

- Landing success in the 90-launch IBM dataset is 66.7% (60/90).
- Success changes substantially over historical years, consistent with an evolving program.
- SVM achieves the highest observed test accuracy in the uploaded run: 83.3% (15/18).

Limitations

- Small and historically bounded dataset; leakage and source differences require care.
- SQL (101 launches) and classification (90 launches) are separate datasets.
- The Folium and Dash HTML exports need static screenshots for stronger slide-level evidence.

Reproducibility and project repository

Repository URL

https://github.com/hajar-hajiri/IBM-Data-Science-Capstone-SpaceX

Contents in the supplied ZIP

- notebooks/01–08: API, wrangling, scraping, SQL, EDA, Folium, Plotly and ML
- data/: cleaned CSVs, SQL result tables, model_comparison.csv
- figures/: EDA figures, SQL chart and model confusion matrices
- dashboard/: interactive HTML exports

Verification note: notebook 06_Folium_Maps.ipynb has no saved execution outputs, even though a map HTML export exists.